

c교를라b도 남교라남기o 남교라교

c교} x 3

Dc교} x 3X prw} t Vt p교} v u교} N교}교교
DM{p교}ur p교} . /

Q교} 교Dw교} 5v교} 4~ 5교} p} v5교} x 3t p교} 3} vx tt 교} v

st bo ?A8

A8A

Upvv{t cx} x 4
bt pq~감 2BC7 7= 4
2 Ptp감만 아 vx} tt 감 v 4
bwp감 BC7 거7= . BC7 거C /

A8B

감만개 s	썩 감	.6G 27G /j l
가{p감	썩 감	.7G7 28G8 29G9 /
감 거	~qy r 감	. p{t 5t p{t/
pvt	ㄴ~p 감	. 7AA /
감 감가	썩 감	5
가 감w	썩 감	5
up 감	ㄴ~p 감	
t qp 감ts	~qy r 감	.M5_5b2 8 /
strz	~qy r 감	. >BB 2AA8+ /

st bo ?B8

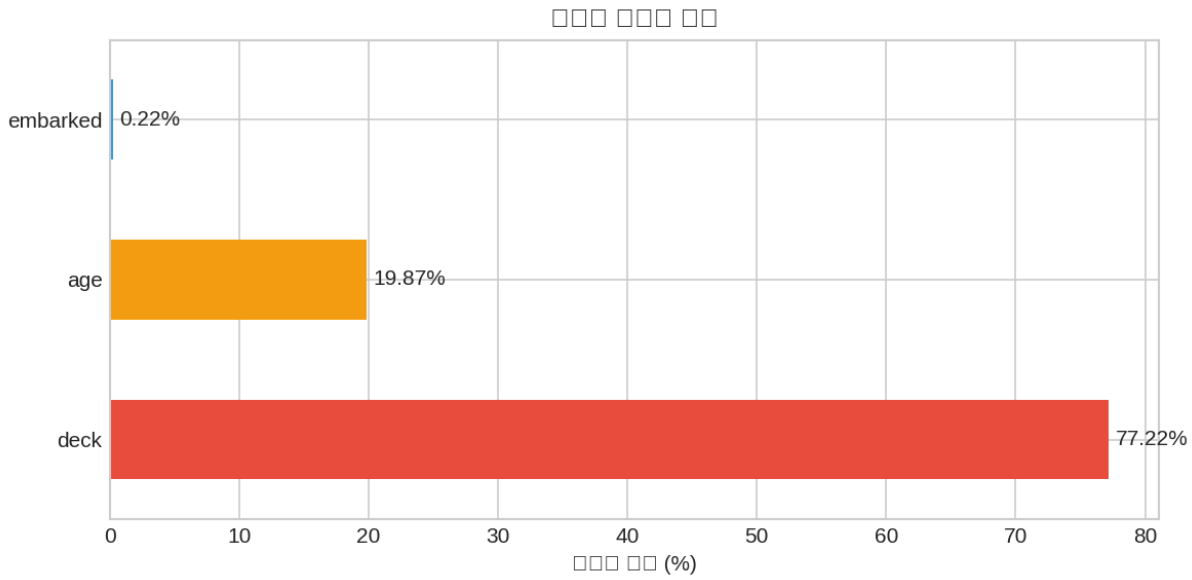
2baT3

B8A

strz.AA8+ /2pvt.7C4+ /2t| qp>2ts.648+ /

4strz

4



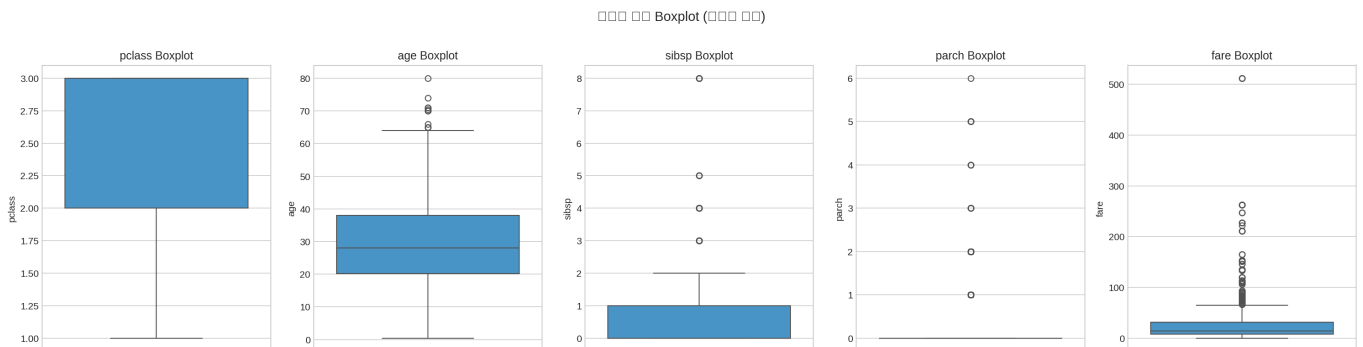
B8B

20드>당>대>형>

S_a

2up>2.7948+ /27p>2w.8947+ /

4



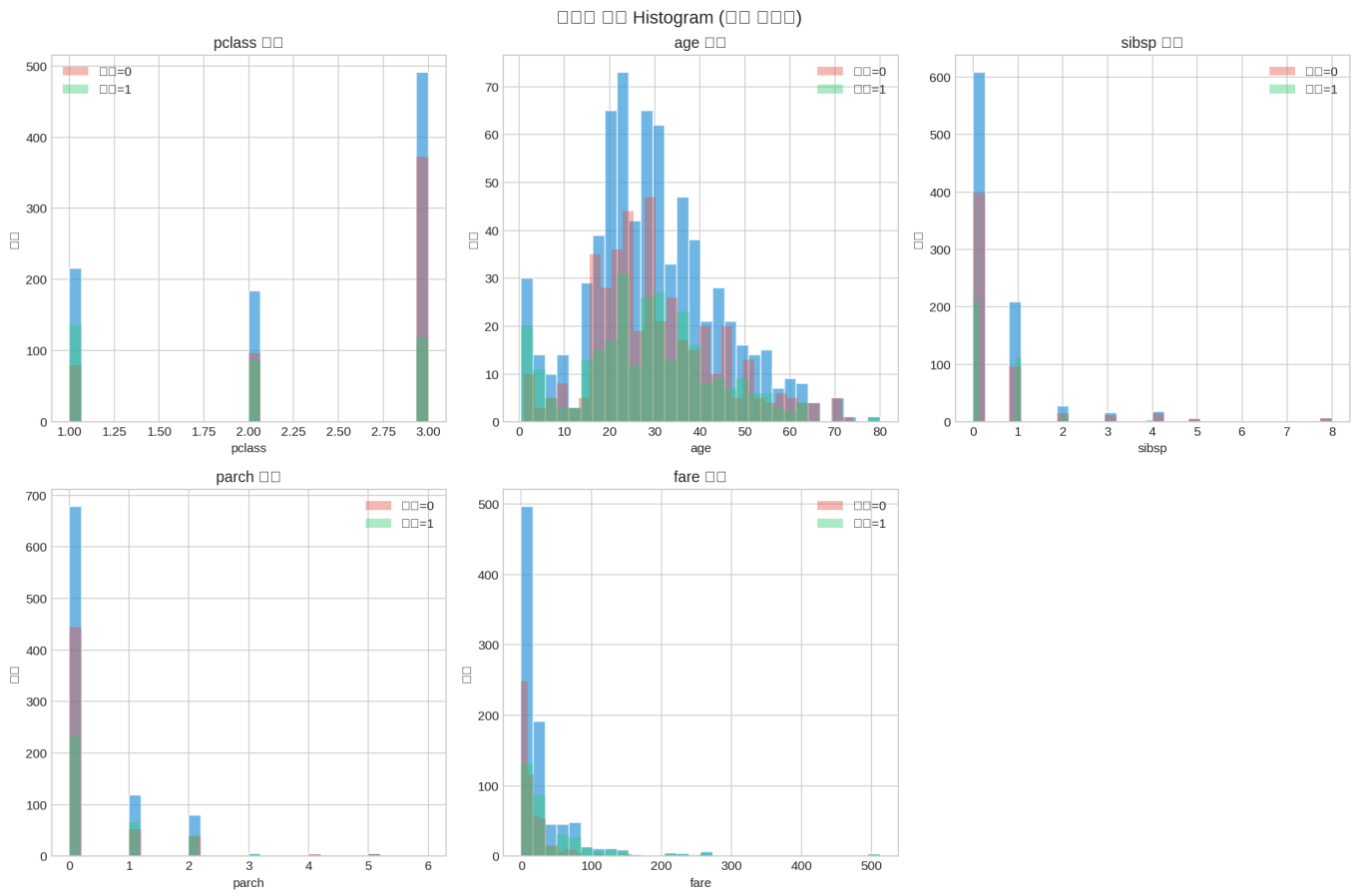
B8C

2e 남뵐 7 가뵐 3

4pvt

2up간

4



.감\6t | p{t/

27 .가{p\감\67/

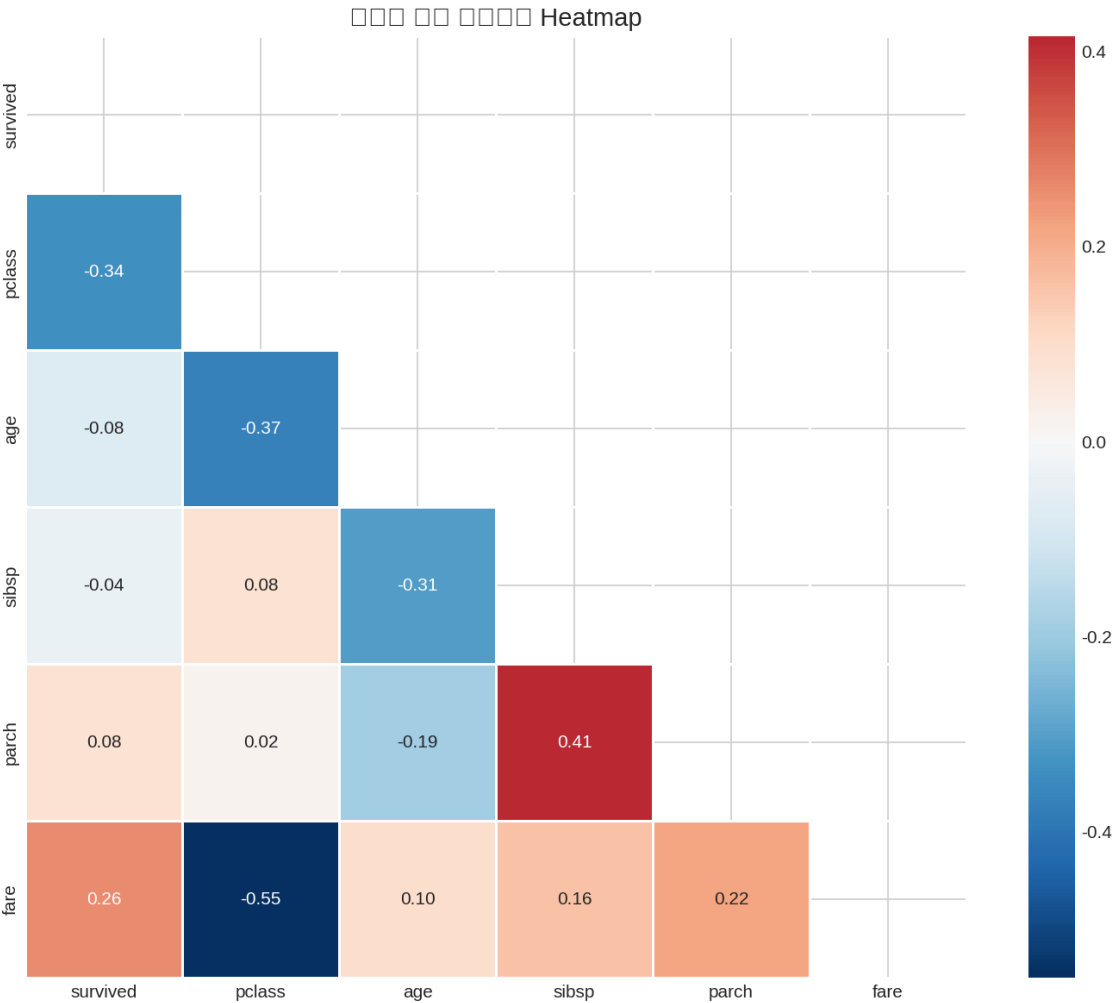
4

Countplot



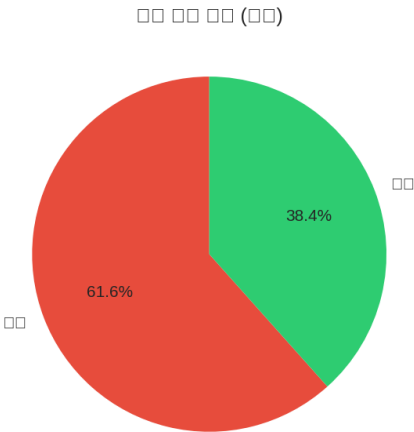
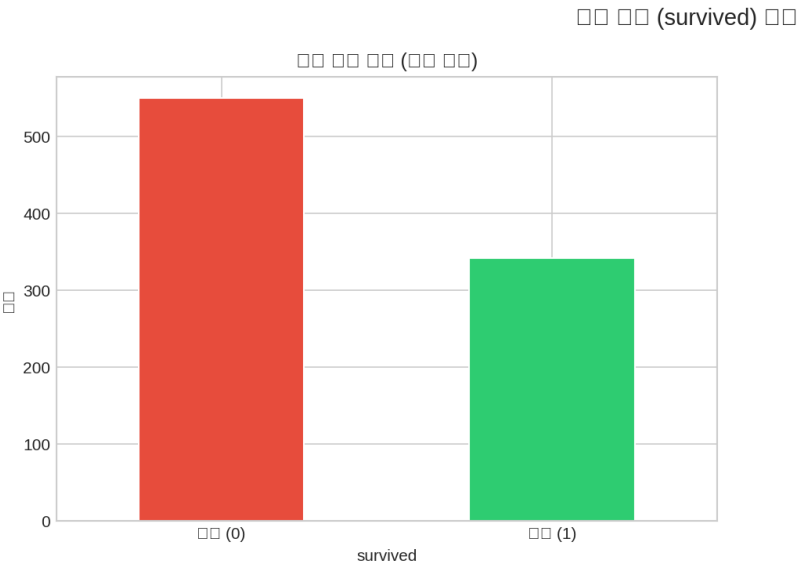
B&F 2e 교재별 고장

.검개개 s/ Dp간.164>/H가{p검340:/4가{p검 up간 .364= / 4



B&F

=: C .>74+ /가함 9: 8 .9B4 + / 4



B8G ba T

j l
 3strzD AA8+
 3pvtD 7C4+ 3H
 3t| qp갈tsD 648+ 3H

j l
 3 >74+ 가람 9B4 + . /

j l
 3갈게 .A: + 가람7C+ /
 3가{p갈7 H8 H9
 3p갈D 3H
 3pvtD .게76 /
 3p갈

st bo ?C8

C8A

= D

7/ up| x간감게 G감q감1 가간w17. /
 8/ xhp{~} t Gup| x간감게 7 7. /
 9/ up간n가 간가 감} Gup간 5up| x간감게 .7 /
 : / pvt nv간감가G .Mwx5ctt} 5h~감} v5X xs s{t 5bt} x~간
 =/ up간nq} G : .V~객5X xs 5Rxw5et 간Rxw/

C8B

: 4N간가 .7AC / 4

X tp}	BC7	6
X tsxp}	BC7	6
X ~감P간감} 감	BC7	6
N간가	A78	6

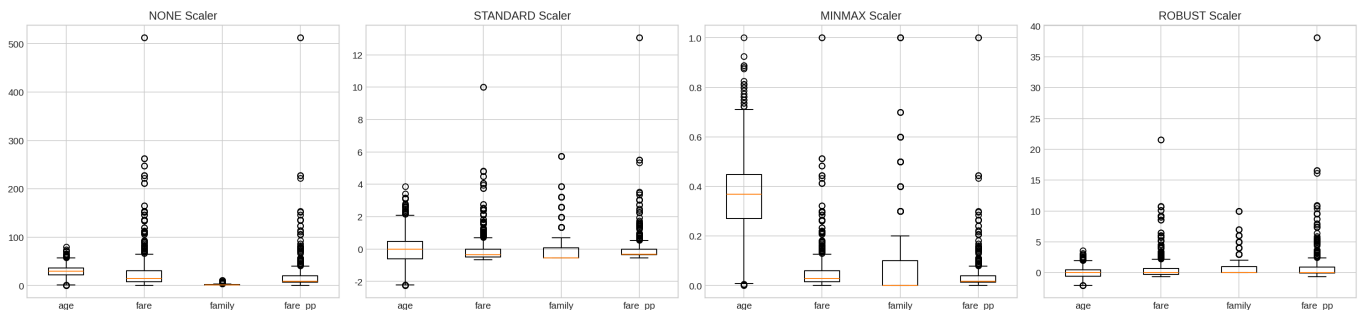
C8C

[] t 3~2D} r~s} vD79 3H7C . 2 /
 Vpqt{ O} r~s} vD79 . 2 /

C8D

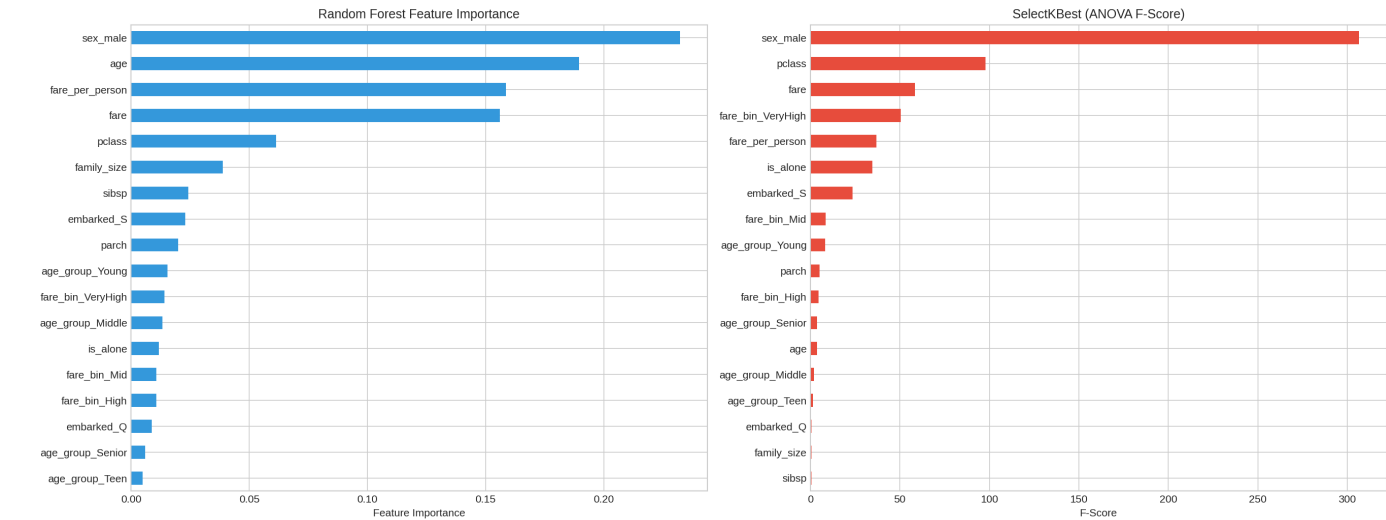
b} s p} brp{t 간X } X p} brp{t 간a~q감brp{t 간 4 up간 a~q감brp{t 간 4

□□□□ □□□ □□ □□



st bo ?D8 2c 교량과 s 교차점도

9 Dap} s~| P~간 값 Ptpglt \$ 가 값} rt 2bt {tr 값 Lt 값 KY[eK P/2aPO



D8A r cb 2t 드잉H3

가{p 값} pvt 2up| x{간 값} 2xhp{~} t 2 값 가| p{t 2pvt nv 간 값} hh~ 값} v2up 간 nq} nR xw 2up 간 nq} net 간 R xw

D8B c 교량과 s 교차점도 :

T 과량과 과하		cA7s 과량과		r nV7TuV
	7B	64A>=	64A767	64B6B8
76	76	64AC99	64A87B	64B79B

76

4

4

st bo ?E8

E8A

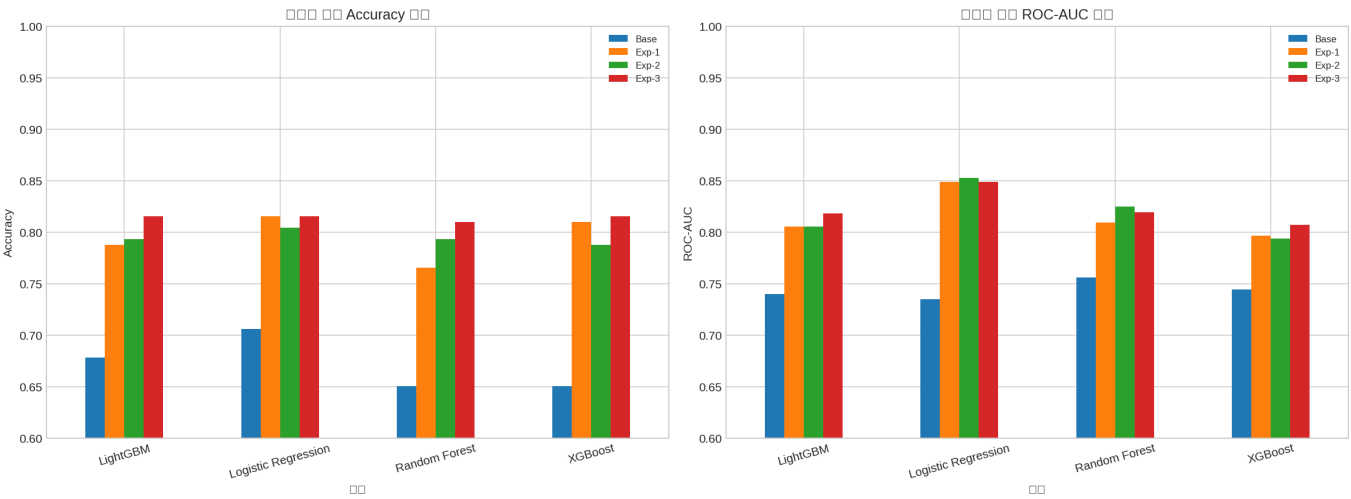
c } } } } }				
Lp}	.N}	.	/	g
O}	Xtp}	[} t3~}	b} sp}	g
O}	Xtsxp}	Vpqt{	X } X p}	[
O}	X ~ } P} } }	[} t3~}	a~q}	[

E8B

T }		o }		r }	cA	r nV7TuV
Lp}	Va	64A6>9	64B7B	647A8	64BB8	64A9=7
Lp}	aP	64=69	64>8=	6486A	64C68	64==C
Lp}	gQL~~}	64=69	64>6>	649AC	64C>B	64A: ::
Lp}	VxwQLX	64AB9	64C>B	649AC	647>A	64A:
O}	Va	64B7=>	64AC69	64A767	64A: B7	64B: BC
O}	aP	64A>=:	64C=A	64C=A	64C=A	64B6C9
O}	gQL~~}	64B767	64AAB	64A767	64A: 8:	64AC>9
O}	VxwQLX	64ABAA	64A9B=	64C=A	64A7>:	64B6==
O}	Va	64B6: =	64AA: 8	64C=A	64A98B	64B=8A
O}	aP	64AC99	64A9=9	64A8: >	64A8CC	64B8: >
O}	gQL~~}	64ABAA	64A: >	64B78	64A787	64AC9A
O}	VxwQLX	64AC99	64A: 8:	64A767	64A8=C	64B6=A
O}	Va	64B7=>	64B	64C=A	64A: : 8	64B: B>
O}	aP	64B767	64A>C8	64A8: >	64A: >9	64B7C:
O}	gQL~~}	64B7=>	64AB78	64A8: >	64A=7C	64B6>C
O}	VxwQLX	64B7=>	64AA8A	64A9C7	64A==>	64B7B8

E8C 2r nV7TuV 3

r nV7TuV		
Lp}	ap} s~ P~ }	64A==C
O}	V~vx} atv }	64B: BC
O}	V~vx} atv }	64B=8A
O}	V~vx} atv }	64B: B>



AJo 농교활교

$$\text{값}\{t\text{값}\} \text{ } x\{t\} \text{ } t \quad M\{\text{값}\} \text{ } c\{t\} \text{ } \text{값}\sim\text{값}\text{ } t\text{값}$$

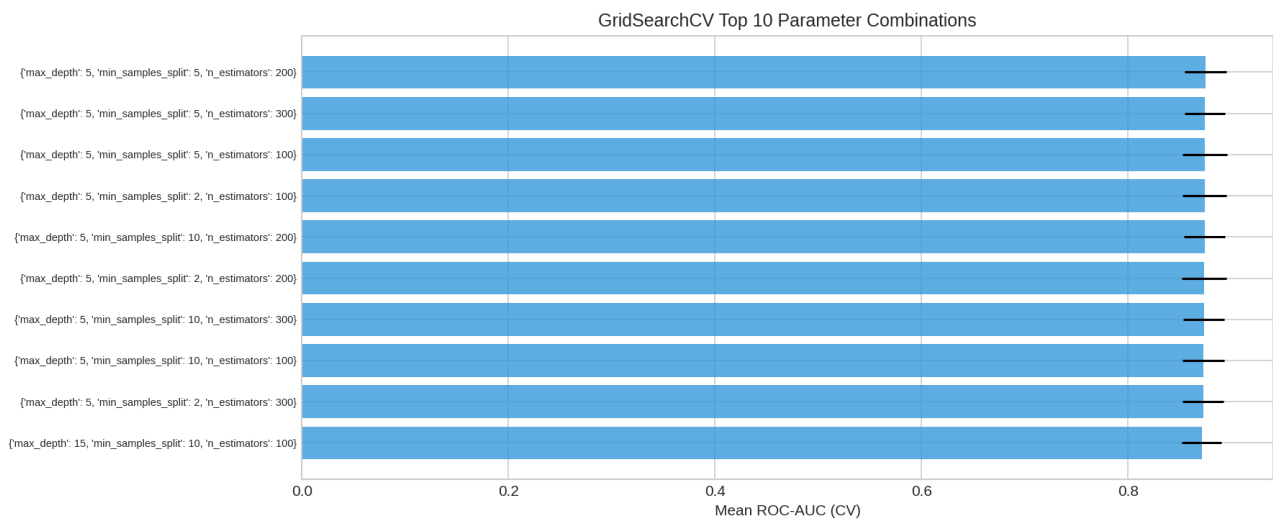
4

DIV~{갑 } c 갑} 감~갑 t 간	DS 기갑간brp{t 간	DS 기갑간[] t R~갑 r~st 갑HM(p갑)ut 간
D	2	2

BJd 레깅스 교과과 Mv

9> D nt 감 감 p 감 감 7662866296612| p 가 h s t 가 감 j = 27627 = 2 ~ } t l 2| x n 감 | 가 t 감 감 가 감 8 = 2761
가 3 ~ { s Me G 7 B 6

D갯 {p갯갯갯 갯} p갯st갯갯D=2-r {p갯갯갯 갯} x n갯 갯 {t갯갯갯 {x갯D=2-r {p갯갯갯 갯} nt갯갯 p갯갯갯D866경
Me a [M3KdMD64BA: C
Krr갯갯D64B767 갯D64A9: : 갯 [M3KdMD64B9BC

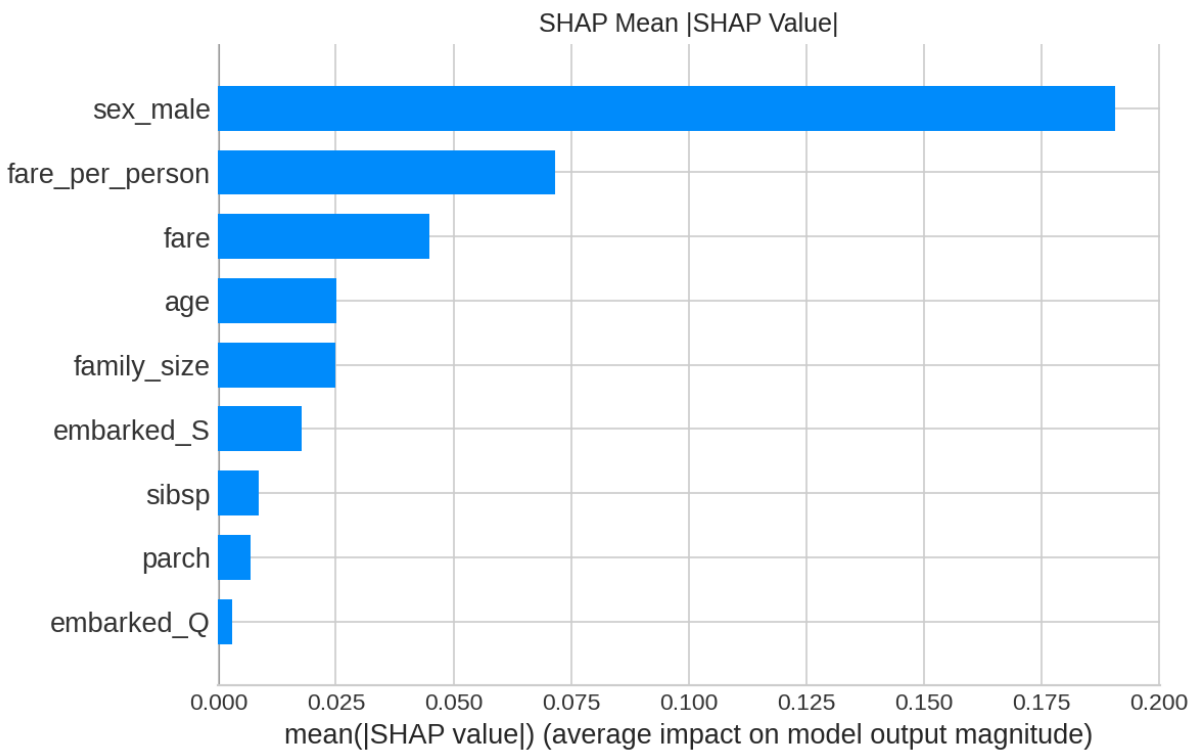
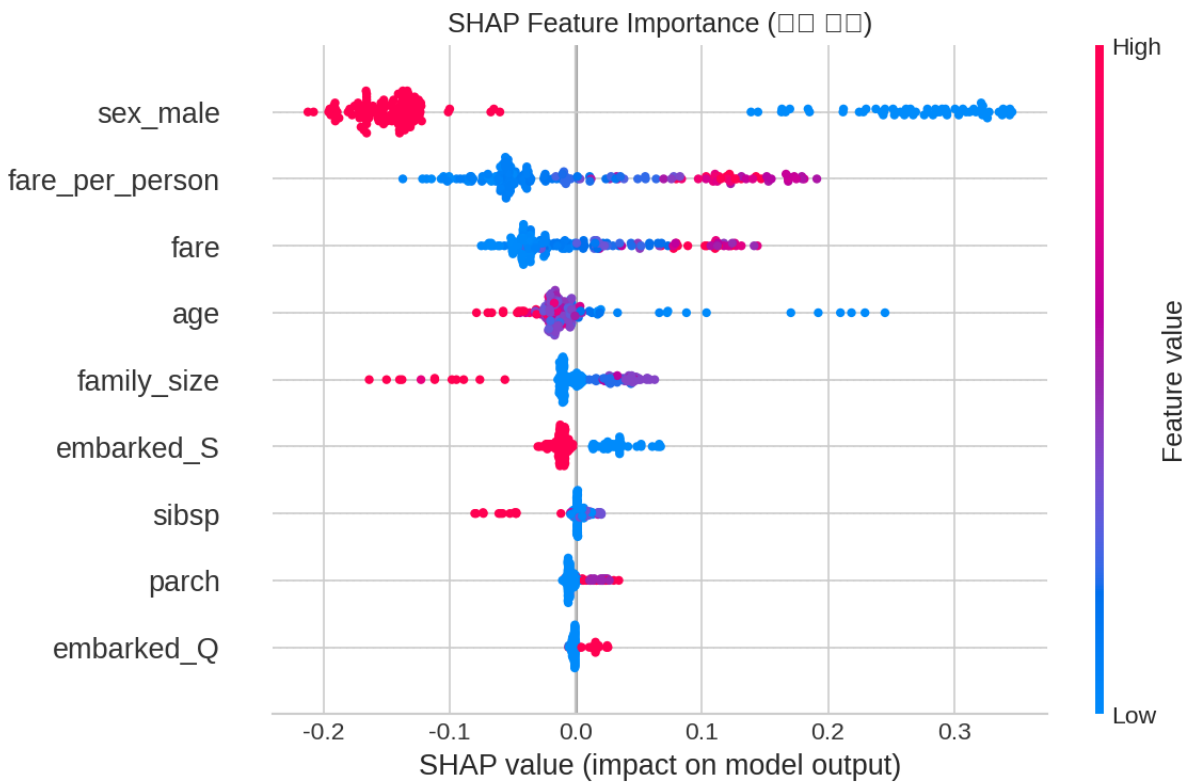


CJseTo

bRK] .bRp가{t 간Kssxg t 거{p} p감} 감
c간tO가{p} t 간

4

4



DJc}교}탈}과}델}당}도}합}도}교}

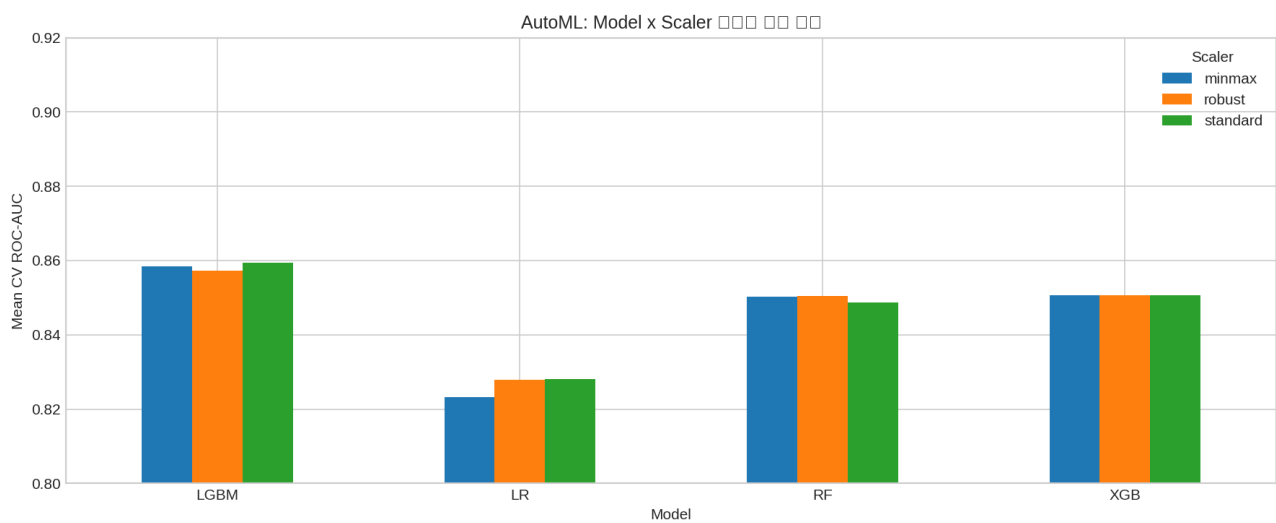
: .Va2aP2gQL ~~~}g}p}w}w}Q}LX/ Ptp}g}p}n} \$ 가}g}p} rt 4



EJT로봇 i

8: . \$ 기gk}k}n 기brp{t}k}n 기X~st{: / 4
 D\$ 기gk}k}n | tsx} 2brp{t}k}n | tsx} spgk} 2X ~st{GVQLX
 Me a[M3KdMD64B>6> .1536468=B/

델 당로와	s 과테라	l 드로테	Vv r n V7TuV	s 란
tsx}	gk} spgk}	VQLX	64B>6>	6468=B
tp}	} p거	VQLX	64B>6=	6468B
tp}	gk} spgk}	VQLX	64B=B8	6468A
tsx}	간qgk}k}n	VQLX	64B=A=	64689:
tp}	간qgk}k}n	VQLX	64B=A	6468>C



A8 R

O[가] B.X ts xp} 5Vpqt{5X [가] X p[가] Pb/ O[가] B.P[가] [가] } tR~[가] a~q[가] Pb/ 4
2Pt p[가] bt{tr[가]}

B8n 도[가] 도[가] 도[가] R

4 [] t3R~[가] O r~s[가] v V~vx[가] atv[가] }. / 2 .aP2 gQL~~[가]
Vxw[가] LX/ Vpqt{O r~s[가] v 4 4

C8c 교[가] 교[가] R

4 7B 76 Krr[가] [가] 64A>=3H64AC992 a[M3KdM 64B6B83H64B79B
4 4

D8 R

3V~vx[가] atv[가] } D 4b[가] sp[가] 5a~q[가]
3ap} s~| P~[가] 5gQL~~[가] 5Vxw[가] LX D
3a~q[가] rp{t [가] [가]

E8c 교[가] 도[가] R

Lp[가]. / Ptp[가] O v[가] tt[가] v a[M3KdM A[가] 6+ 4
.Lp[가] 64A==C3HO[가] B 64B=8A/

up| [가] [가] 2[가] p{~} t2up[가] n[가] [가] 2bRK]
4

나름:기환공과델:라함고기 람[가] 도[가] 구교[가] 교[가] 남교[가]